

Linux Containers Lab Guide with Command Explanations and Screenshots

Step 1 - Pull Ubuntu Image

Command: `docker pull ubuntu:24.04`

Explanation: Downloads the Ubuntu Linux image from Docker Hub to your local machine. Images are templates used to create containers.

Expected Output:

```
laxmanp@W4N00TW13C-laxmanp Linux1 % docker pull ubuntu:24.04
24.04: Pulling from library/ubuntu
532a4cf7e47b: Download complete
Digest: sha256:786a8b558f7be160c6c8c4a54f9a57274f3b4fb1491cf65146521ae77ff1dc54
Status: Downloaded newer image for ubuntu:24.04
docker.io/library/ubuntu:24.04

What's next:
  View a summary of image vulnerabilities and recommendations → docker scout quickview ubuntu:24.04
laxmanp@W4N00TW13C-laxmanp Linux1 %
```

Step 2 - Verify Image

Command: `docker images`

Explanation: Displays all images stored locally on your computer.

Expected Output: _____

```
laxmanp@W4N00TW13C-laxmanp Linux1 % docker images
View a summary of image vulnerabilities and recommendations → docker scout quickview ubuntu:24.04
laxmanp@W4N00TW13C-laxmanp Linux1 % docker images

```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
compose-postgres-api-lab-api:latest	309a53ba5531	387MB	97MB	U
dpape/pgadmin4:latest	cefc4cc6b7d9	765MB	175MB	U
nginx:latest	608a100c7165	259MB	64.3MB	U
postgres:16	081f1bc7bd5e	663MB	165MB	U
postgres:latest	29ee7bb30d80	671MB	167MB	U
ubuntu:24.04	786a8b558f7b	141MB	30.8MB	U

```
laxmanp@W4N00TW13C-laxmanp Linux1 %
```

Step 3 - Start Ubuntu Container

Command: `docker run -it ubuntu:24.04 bash`

Explanation: Creates and starts a container from the Ubuntu image and opens an interactive Bash shell.

Expected Output: _____

```
laxmanp@W4N0TW13C-laxmanp Linux1 % docker run -it ubuntu:24.04 bash
root@05e51fe878ca:/#
```

Step 4 - Check Current Directory

Command: pwd

Explanation: Prints the current working directory.

Expected Output: _____

```
laxmanp@W4N0TW13C-laxmanp Linux1 % docker run -it ubuntu:24.04 bash
root@05e51fe878ca:/# pwd
/
root@05e51fe878ca:/#
```

Step 5 - List Files

Command: ls -la

Explanation: Displays all files and folders including hidden files.

Expected Output: _____

```
laxmanp@W4N0TW13C-laxmanp Linux1 % docker run -it ubuntu:24.04 bash
root@05e51fe878ca:/# pwd
/
root@05e51fe878ca:/# ls -la
total 56
drwxr-xr-x  1 root root 4096 Jun 22 04:21 .
drwxr-xr-x  1 root root 4096 Jun 22 04:21 ..
-rwxr-xr-x  1 root root   0 Jun 22 04:21 .dockerenv
lrwxrwxrwx  1 root root   7 Apr 22 2024 bin -> usr/bin
drwxr-xr-x  2 root root 4096 Apr 22 2024 boot
drwxr-xr-x  5 root root 360 Jun 22 04:21 dev
drwxr-xr-x  1 root root 4096 Jun 22 04:21 etc
drwxr-xr-x  3 root root 4096 May  9 14:23 home
lrwxrwxrwx  1 root root   7 Apr 22 2024 lib -> usr/lib
drwxr-xr-x  2 root root 4096 May  9 14:17 media
drwxr-xr-x  2 root root 4096 May  9 14:17 mnt
drwxr-xr-x  2 root root 4096 May  9 14:17 opt
dr-xr-xr-x 213 root root   0 Jun 22 04:21 proc
drwx----- 2 root root 4096 May  9 14:23 root
drwxr-xr-x  4 root root 4096 May  9 14:23 run
lrwxrwxrwx  1 root root   8 Apr 22 2024 sbin -> usr/sbin
drwxr-xr-x  2 root root 4096 May  9 14:17 srv
dr-xr-xr-x 11 root root   0 Jun 22 04:21 sys
drwxrwxrwt  2 root root 4096 May  9 14:23 tmp
drwxr-xr-x 11 root root 4096 May  9 14:17 usr
drwxr-xr-x 11 root root 4096 May  9 14:23 var
root@05e51fe878ca:/#
```

Step 6 - View Linux Version

Command: `cat /etc/os-release`

Explanation: Displays operating system information.

Expected Output: _____

```
drwxr-xr-x 11 root root 4096 May  9 14:23 var
root@05e51fe878ca:/# cat /etc/os-release
PRETTY_NAME="Ubuntu 24.04.4 LTS"
NAME="Ubuntu"
VERSION_ID="24.04"
VERSION="24.04.4 LTS (Noble Numbat)"
VERSION_CODENAME=noble
ID=ubuntu
ID_LIKE=debian
HOME_URL="https://www.ubuntu.com/"
SUPPORT_URL="https://help.ubuntu.com/"
BUG_REPORT_URL="https://bugs.launchpad.net/ubuntu/"
PRIVACY_POLICY_URL="https://www.ubuntu.com/legal/terms-and-policies/privacy-policy"
UBUNTU_CODENAME=noble
LOGO=ubuntu-logo
root@05e51fe878ca:/#
```

Step 7 - Check Current User

Command: `whoami`

Explanation: Displays the currently logged-in user.

Expected Output: _____

```
root@05e51fe878ca:/# whoami
root
root@05e51fe878ca:/#
```

Step 8 - Create File

Command: `touch notes.txt`

Explanation: Creates an empty file named notes.txt.

Expected Output: _____

```
root@05e51fe878ca:/# touch notes.txt
root@05e51fe878ca:/# ls
bin boot dev etc home lib media mnt notes.txt opt proc root run sbin srv sys tmp usr var
root@05e51fe878ca:/#
```

Step 9 - Write Data

Command: `echo "Hello Docker" > notes.txt`

Explanation: Writes text into a file.

Expected Output: _____

```
root@05e51fe878ca:/# echo "Hello Docker" > notes.txt
root@05e51fe878ca:/# cat notes.txt
Hello Docker
root@05e51fe878ca:/#
```

Step 10 - Create Folder

Command: `mkdir training`

Explanation: Creates a new directory.

Expected Output: _____

```
root@05e51fe878ca:/# mkdir training
root@05e51fe878ca:/# ls
bin boot dev etc home lib media mnt notes.txt opt proc root run sbin srv sys tmp training usr var
root@05e51fe878ca:/#
```

Step 11 - View Processes

Command: `ps -ef`

Explanation: Displays running processes inside the container.

Expected Output: _____

```
root@05e51fe878ca:/# ps -ef
UID          PID  PPID  C  STIME TTY          TIME CMD
root         1      0  0  04:21 pts/0    00:00:00 bash
root        17      1  0  04:39 pts/0    00:00:00 ps -ef
root@05e51fe878ca:/#
```

Step 12 - Install Software

Commands: `apt update` && `apt install curl -y`

Explanation: Updates package repositories and installs curl.

Expected Output: _____

```

Setting up libnftnl2:arm64 (1.0.3-1ubuntu0.3) ...
Setting up krb5-locales (1.20.1-6ubuntu2.6) ...
Setting up libldap-common (2.6.10+dfsg-0ubuntu0.24.04.1) ...
Setting up libkrb5support0:arm64 (1.20.1-6ubuntu2.6) ...
Setting up libsasl2-modules-db:arm64 (2.1.28+dfsg1-5ubuntu3.1) ...
Setting up librtmp1:arm64 (2.4+20151223.gitfa8646d.1-2build7) ...
Setting up libk5crypto3:arm64 (1.20.1-6ubuntu2.6) ...
Setting up libsasl2-2:arm64 (2.1.28+dfsg1-5ubuntu3.1) ...
Setting up libkrb5-3:arm64 (1.20.1-6ubuntu2.6) ...
Setting up openssl (3.0.13-0ubuntu3.11) ...
Setting up publicsuffix (20231001.0357-0.1) ...
Setting up libldap2:arm64 (2.6.10+dfsg-0ubuntu0.24.04.1) ...
Setting up ca-certificates (20260601~24.04.1) ...
debconf: unable to initialize frontend: Dialog
debconf: (No usable dialog-like program is installed, so the dialog based frontend cannot be used. at /usr/share/perl5/Debconf/FrontEnd/Di
alog.pm line 79.)
debconf: falling back to frontend: Readline
debconf: unable to initialize frontend: Readline
debconf: (Can't locate Term/ReadLine.pm in @INC (you may need to install the Term::ReadLine module) (@INC entries checked: /etc/perl /usr/
local/lib/aarch64-linux-gnu/perl/5.38.2 /usr/local/share/perl/5.38.2 /usr/lib/aarch64-linux-gnu/perl5/5.38 /usr/share/perl5 /usr/lib/aarch
64-linux-gnu/perl-base /usr/lib/aarch64-linux-gnu/perl/5.38 /usr/share/perl/5.38 /usr/local/lib/site_perl) at /usr/share/perl5/Debconf/Fro
ntEnd/Readline.pm line 8.)
debconf: falling back to frontend: Teletype
Updating certificates in /etc/ssl/certs...
121 added, 0 removed; done.
Setting up libgssapi-krb5-2:arm64 (1.20.1-6ubuntu2.6) ...
Setting up libssh-4:arm64 (0.10.6-2ubuntu0.4) ...
Setting up libcurl4t64:arm64 (8.5.0-2ubuntu10.9) ...
Setting up curl (8.5.0-2ubuntu10.9) ...
Processing triggers for libc-bin (2.39-0ubuntu8.7) ...
Processing triggers for ca-certificates (20260601~24.04.1) ...
Updating certificates in /etc/ssl/certs...
0 added, 0 removed; done.
Running hooks in /etc/ca-certificates/update.d...
done.
root@05e51fe878ca:/#

```

Step 13 - Networking

Commands: hostname, hostname -I

Explanation: Displays hostname and IP address.

Expected Output: _____

```

0 added, 0 removed; done.
Running hooks in /etc/ca-certificates/update.d...
done.
root@05e51fe878ca:/# hostname
05e51fe878ca
root@05e51fe878ca:/# hostname -I
172.17.0.2
root@05e51fe878ca:/#

```

Step 14 - Create User

Command: useradd student

Explanation: Creates a new Linux user.

Expected Output: _____

```

root@05e51fe878ca:/# useradd student
root@05e51fe878ca:/#

```

Step 15 - Permissions

Command: `chmod 777 secret.txt`

Explanation: Changes file permissions to full access.

Expected Output: _____

```
root@05e51fe878ca:/# ls -l notes.txt
-rw-r--r-- 1 root root 13 Jun 22 04:27 notes.txt
root@05e51fe878ca:/# chmod 777 notes.txt
root@05e51fe878ca:/# ls -l notes.txt
-rwxrwxrwx 1 root root 13 Jun 22 04:27 notes.txt
root@05e51fe878ca:/#
```

Step 16 - Environment Variables

Command: export COMPANY=Training

Explanation: Creates an environment variable.

Expected Output: _____

```
root@05e51fe878ca:/# export COMPANY=Training
root@05e51fe878ca:/# env
HOSTNAME=05e51fe878ca
COMPANY=Training
PWD=/
HOME=/root
LS_COLORS=rs=0:di=01;34:ln=01;36:mh=00:pi=40;33:so=01;35:do=01;35:bd=40;33;01:cd=40;33;01:or=40;31;01:mi=00:su=37;42:ow=34;42:st=37;44:ex=01;32:*.tar=01;31:*.tgz=01;31:*.arc=01;31:*.arj=01;31:*.taz=01;31:*.lha=01;31:*.lzh=01;31:*.flr=01;31:*.ttx=01;31:*.tzo=01;31:*.t7z=01;31:*.zip=01;31:*.z=01;31:*.dz=01;31:*.gz=01;31:*.lcz=01;31:*.l7z=01;31:*
```

Step 17 - Exit Container

Command: exit

Explanation: Leaves the container shell.

Expected Output: _____

```
root@05e51fe878ca:/# exit
exit
laxmanp@W4N00TW13C-laxmanp Linux1 %
```